

ET AI HACKATHON 2026

# AXIS

## Unified Asset & Operations Brain

Problem Statement 8: AI for Industrial Knowledge Intelligence

*Detailed Submission Document*

**Team AXIS**

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## 1. Executive Summary

AXIS is a Unified Asset & Operations Brain for asset-intensive industries. It ingests heterogeneous plant documents — P&IDs, CMMS work orders, SOPs, inspection reports, regulatory standards and near-miss reports — into a single, continuously-updated knowledge layer, then exposes that knowledge through a conversational Expert Copilot and cooperating intelligence agents for maintenance root-cause analysis and regulatory compliance.

Every answer AXIS gives is grounded in a specific, cited source document, and every cross-functional insight it surfaces is traceable back to the records that produced it. The result is a system that turns fragmented institutional knowledge into a queryable asset, available at the point of need.

## 2. Problem Statement

Asset-intensive industrial plants operate across many disconnected document systems. Engineering drawings, maintenance records, safety procedures, inspection reports and regulatory submissions live in separate silos, and the people who could connect them across a plant's history are retiring faster than that knowledge is being captured.

|                                                                                                 |                                                                                           |                                                                               |                                                                                           |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>35%</b><br>of working hours spent searching for information or recreating existing documents | <b>7-12</b><br>disconnected document systems in the average large Indian industrial plant | <b>18-22%</b><br>of unplanned downtime linked to incomplete equipment history | <b>25%</b><br>of India's experienced industrial engineers retiring within the next decade |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

Figure 1. Scale of the knowledge-fragmentation problem in Indian heavy industry.

This is not a filing problem. Knowledge fragmentation directly costs safety margin, quality consistency, and operational uptime — and it compounds every year experienced staff leave without a system that has captured what they know.

## 3. Solution Overview

AXIS addresses the challenge through four cooperating layers:

- Ingestion & Extraction — OCR, computer vision for P&ID parsing, and named-entity recognition turn scanned drawings and forms into structured, linked entities (equipment tags, process parameters, standards, personnel, dates).
- Unified Knowledge Layer — a knowledge graph maintains relationships across every document type, backed by a vector index for semantic retrieval beyond exact keyword match.
- Expert Knowledge Copilot — a retrieval-augmented conversational interface that answers operational, maintenance and engineering questions with source citations and a confidence score on every answer, built mobile-first for field technicians.
- Agentic Intelligence — a Maintenance & RCA agent, a Compliance & Quality agent, and a Lessons Learned engine that proactively cross-reference records no single reviewer would connect on their own.

## 4. System Architecture

AXIS follows a six-layer pipeline: Sources -> Ingestion -> Knowledge Layer -> Agentic Intelligence -> Experience Layer -> Users. New documents, incidents and inspections continuously re-enter the ingestion layer, so the knowledge graph never goes stale.

*See the accompanying architecture diagram (submitted separately / included in the GitHub repository under docs/) for the full visual layout of this pipeline.*

## 5. Key Capabilities (Prototype)

### 5.1 Knowledge Corpus & Entity Linking

The working prototype ingests six representative document types — a P&ID, a CMMS work order, an SOP, an inspection report, a regulatory standard excerpt (OISD-STD-118), and a near-miss report — and resolves them onto a shared set of equipment tags and standards, visualized as a live, interactive knowledge graph.

### 5.2 Expert Knowledge Copilot

A user can ask, in plain language, e.g. “What’s the maintenance and incident history for P-207A?”. The Copilot retrieves the relevant documents, synthesizes an answer strictly from that retrieved material, and returns the answer together with the exact source document IDs and a confidence score — nothing is generated outside the grounded context.

### 5.3 Cross-Functional Signals

The prototype demonstrates the RCA and Compliance tracks concretely:

- RCA Agent — links three bearing replacements on Pump P-207A across 14 months to a previously-silenced vibration alarm, flagging a predictive maintenance and lubrication audit that no single work order would have surfaced.
- Compliance Agent — cross-references Vessel V-112’s corrosion rate (3.9x the site average) against OISD-STD-118’s risk-based inspection interval requirement, confirming the shortened interval and flagging the deviation basis needed for audit evidence.

### 5.4 Live Knowledge Graph

As the Copilot answers a query, the knowledge graph panel highlights the exact documents and entities it drew from — making the system’s reasoning visible and auditable rather than a black box.

## 6. Technology Stack

| Layer             | Approach in this prototype | Production-scale direction                                     |
|-------------------|----------------------------|----------------------------------------------------------------|
| Ingestion         | Structured mock corpus     | OCR / Document Intelligence + Computer Vision for P&ID parsing |
| Entity Extraction | Manually tagged entities   | NER pipeline for equipment tags,                               |

| Layer           | Approach in this prototype            | Production-scale direction                                          |
|-----------------|---------------------------------------|---------------------------------------------------------------------|
|                 |                                       | standards, personnel, dates                                         |
| Knowledge Layer | In-browser entity graph               | Graph database (e.g. Neo4j) + vector index (e.g. pgvector / Chroma) |
| Retrieval       | Lexical + extractive scoring, offline | Embedding-based semantic retrieval over the vector index            |
| Copilot         | Client-side extractive synthesis      | LLM-backed generation via a server-held API key                     |
| Agents          | Rule-based demo signals               | Scheduled/event-driven agents over the live knowledge graph         |
| Interface       | Responsive web dashboard              | Same dashboard, extended to native mobile for field use             |

## 7. Business Impact & Judging Criteria Alignment

| Criterion            | Weight | How AXIS delivers                                                                                                                    |
|----------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| Innovation           | 25%    | Cross-functional agents surface connections (maintenance <-> near-miss, inspection <-> regulation) that single-purpose tools do not. |
| Business Impact      | 25%    | Directly targets the 35% search-time drain and the 18–22% downtime tied to fragmented records.                                       |
| Technical Excellence | 20%    | Every answer is grounded with mandatory citations and a confidence score — no ungrounded generation.                                 |
| Scalability          | 15%    | Ingestion, the knowledge layer, and each agent are designed as independently scalable services.                                      |
| User Experience      | 15%    | One conversational interface, mobile-ready, replacing seven to twelve separate systems.                                              |

## 8. Scalability & Future Roadmap

- Replace the offline extractive Copilot with a server-hosted LLM call, keeping the API key off the client and enabling proper query logging.
- Swap keyword retrieval for a real embedding-based vector index for semantic search at corpus scale.
- Add computer-vision P&ID parsing to ingest engineering drawings directly rather than as pre-tagged entities.
- Extend agents from demonstration signals to scheduled, event-driven services running continuously against the live graph.

- Add a document upload flow so the knowledge graph updates in real time as new work orders, inspections and incidents are filed — directly addressing the challenge's “continuously updated” requirement.

## 9. Submission Links

- GitHub Repository: [add your repository URL here]
- Live Prototype: included in the GitHub repository (industrial-knowledge-brain.html) — open directly in any browser.
- Demo Video: submitted alongside this document per the submission form.